

Campus Grievance Redressal & Maintenance Tracker

Hackathon Project Report

Abstract

Campus maintenance complaints often go unresolved because they are reported through informal channels such as WhatsApp, phone calls, or verbal communication. Without a centralized tracking mechanism, complaints are easily forgotten, responsibilities remain unclear, and students receive no updates regarding the status of their issues.

The Campus Grievance Redressal & Maintenance Tracker addresses this challenge by providing a transparent complaint management platform where every issue is tracked until closure. The system introduces ownership, accountability, escalation mechanisms, and service-level timers to ensure timely resolution.

Problem Statement

Educational institutions receive numerous maintenance and grievance requests every day, including broken electrical equipment, plumbing issues, damaged furniture, internet connectivity problems, and hostel maintenance.

The current reporting process suffers from several limitations:

- Complaints are scattered across multiple communication channels.
- No centralized complaint database exists.
- Responsibility is not clearly assigned.
- Students cannot track complaint progress.
- Delayed responses reduce trust in the system.

These issues lead to inefficiency, poor communication, and unnecessary delays in resolving campus problems.

Objectives

The project aims to:

- Centralize complaint management.
- Improve transparency.
- Assign ownership for every complaint.
- Enable real-time status tracking.
- Reduce complaint resolution time.
- Automatically escalate delayed complaints.
- Increase student satisfaction.

Proposed Solution

The proposed system offers an integrated complaint management platform.

Students can submit complaints along with supporting images and select the appropriate category.

Administrators receive the complaint immediately, assign it to the concerned department, monitor the resolution progress, and close the complaint after successful completion.

The platform ensures complete visibility for both students and administrators.

Key Features

Student Module

- User Login
- Complaint Registration
- Attachment Upload
- Complaint Categories
- Status Tracking
- Complaint History

- Notifications

Admin Module

- Dashboard
- Complaint Assignment
- Status Management
- SLA Monitoring
- Escalation Rules
- Reports & Analytics

System Workflow

Student Registers Complaint



Complaint Stored in Database



Admin Reviews Complaint



Assigned to Responsible Department



Issue Resolution



Status Updated



Student Notified



Complaint Closed

Technology Stack

Frontend

- React (Vite)
- Axios
- React Context API
- Recharts
- Custom CSS (Light/Dark Theme)
- Vite Preview

Backend

- Flask (Python)
- Flask-SQLAlchemy
- Flask-JWT-Extended (JWT Authentication)
- Flask-CORS
- Werkzeug Security (Password Hashing)
- python-dotenv
- Gunicorn

Database

- SQLite

Architecture

- Monolithic deployment(Flask serves React build)

- RESTful API
- Role - Based Access Control(Student/Admin)

Deployment

- Render (Free Tier) with Github Auto-Deploy

Expected Outcomes

The system will:

- Increase transparency.
- Improve accountability.
- Reduce complaint resolution time.
- Maintain complete complaint history.
- Improve communication between students and administration.

Future Enhancements

- AI-powered complaint prioritization.
- Mobile application.
- QR-code based complaint registration.
- Email and SMS notifications.
- Predictive maintenance dashboard.

Team Members

Team Lead

-Kripa

Frontend Developer

-Akash

Backend Developer

-Kripa

Documentation & GitHub

-Saumya & Sristi

Presentation

-Avinash

Conclusion

The Campus Grievance Redressal & Maintenance Tracker transforms the traditional complaint handling process into a transparent and accountable digital platform. By introducing centralized complaint tracking, department assignment, escalation rules, and real-time status monitoring, the system ensures that every grievance is addressed efficiently. This solution strengthens trust between students and the institution while promoting timely maintenance and better campus management.